TDWG Standard · GBIF & OBIS · Open Biodiversity Data

Darwin Core (DwC), Data Publishing, Citation and IPT

Mobilizing, standardizing, and crediting global biodiversity records across the Open Science research lifecycle.

Begin Lecture →

Open Science Course

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The Problem: The Biodiversity Tower of Babel

Before standardization, natural history collections and field scientists recorded data in disparate silos:

🏷️ Taxonomy Chaos

Synonyms, typos, regional vernacular names, and conflicting authorities without persistent identifiers.

🗺️ Spatial Ambiguity

"Behind the old mill" vs degrees/minutes/seconds vs local UTM grids; lack of coordinate uncertainty metrics.

📆 Temporal Inconsistency

`04/05/06` (May 4th or April 5th?), arbitrary date intervals, time zones missing.

🎯 **The Consequence:** Global synthesis (climate change range shifts, extinction tracking, invasive species modeling) was virtually impossible without manual data wrangling.

What is Darwin Core (DwC)?

Darwin Core (DwC) is the international standard maintained by **TDWG (Biodiversity Information Standards)**.

- **A Glossary of Terms:** A stable vocabulary with precise URI definitions (e.g. `dwc:occurrenceID`, `dwc:scientificName`).
- **Pragmatic & Flexible:** Storage-independent (flat CSV, relational SQL, JSON-LD, RDF XML).
- **The Lingua Franca of Global Portals:** The foundational exchange format for **GBIF**, **OBIS**, and national biodiversity nodes.

🔗 **Essential Reference Guide:**
Consult the official **Darwin Core Quick Reference Guide (dwc.tdwg.org)** for every term definition, namespace, and normative examples.

Official TDWG Identity tdwg.org ↗

Standard:	**Biodiversity Information Standards (TDWG)**
Ratified:	October 2009 (Living standard)
Guide URL:	dwc.tdwg.org
Namespace:	http://rs.tdwg.org/dwc/terms/
Quick List:	dwc.tdwg.org/list/

"Darwin Core is primarily based on taxa, their occurrence in nature as documented by observations, specimens, samples, and related information."

What is GBIF?

The **Global Biodiversity Information Facility** (GBIF.org) is an international network and research infrastructure funded by governments worldwide.

🌐 The World's Largest Biodiversity Aggregator

GBIF provides free and open access to biodiversity data globally, mobilizing evidence from museum specimens, ecological surveys, eDNA, and citizen science (e.g. iNaturalist, eBird).

📜 **Global Open Mandate:** Standardized open licenses (CC0, CC-BY, CC-BY-NC) and citable DOIs for all datasets and search queries.

🧬 **Powered by Darwin Core:** Universal lingua franca harmonizing disparate records into a single queryable global index.

🌍 **Conservation & Policy:** Essential evidence for IUCN Red List assessments, IPBES reports, and biodiversity targets.

2.9+ Billion

Species Occurrence Records

Museum specimens, citizen science, eDNA, sensors

2,500+

Publishing Institutions

Universities, museums, government agencies, NGOs

100+

Voting Participants & Nodes

Country delegations and international organizations

How GBIF Operates: Nodes & Governance

GBIF is **not an uncurated upload bucket** — it is a **federated, endorsed network**:



Country / Participant Nodes

Designated national teams (e.g. GBIF Norway, GBIF Spain) serving as the official bridge between local institutions and the global infrastructure.

Institutional Endorsement

Strict vetting: Node managers must review and approve every publishing organization before its datasets can be registered on GBIF.

Federated Publishing (IPT)

Organizations keep ownership of their data. The node hosts or assists with the **IPT**, which exports Darwin Core Archives for GBIF to harvest.

GBIF Dataset Classes Overview

GBIF standardizes biodiversity publishing into **4 distinct dataset classes** (gbif.org/dataset-classes):

1. Occurrence

`Core: dwc:Occurrence`

Evidence of a species at a given time and place (museum specimens, citizen science observations, camera traps).

`Most common on GBIF`

2. Sampling-Event

`Core: dwc:Event + Occ Ext`

Standardized environmental monitoring, surveys, transects with quantifiable effort and non-detection (absence) data.

`Ecological time-series`

3. Checklist

`Core: dwc:Taxon`

Catalogues of taxa, national inventories, red lists, and taxonomic classifications without specific coordinate events.

`Taxonomic backbone`

4. Metadata-only

`Core: None (EML XML)`

Describing undigitized physical collections, historical literature, or projects awaiting electronic mobilization.

`Collection discovery`

GBIF Recommendations: Mandatory vs. Recommended Terms

Explore terms per dataset class based on gbif.org/dataset-classes and dwc.tdwg.org:

Occurrence | Sampling-Event | Checklist | Metadata-Only

All | ● Mandatory Only

Occurrence Datasets (Core: dwc:Occurrence)

GBIF Class Docs

Evidence of the presence, abundance, or absence of a biological organism at a specific location and date.

Darwin Core Term	GBIF Status	Definition & Purpose	Example Value
dwc:occurrenceID Occurrence	● Mandatory	Globally unique, persistent identifier for the occurrence record. ↳ Primary key; GBIF indexing fails without it	urn:uuid:8b3e8e2b-2e92-498c-851c-a1288c8eb25a
dwc:basisOfRecord Occurrence	● Mandatory	Controlled vocabulary denoting the specific nature of the record. ↳ Required controlled vocabulary for filtering	HumanObservation, PreservedSpecimen, MaterialSample

The Anatomy of a Darwin Core Record

A practical example of how a biological observation maps directly to standardized DwC terms:

```json
{
  "dwc:occurrenceID": "urn:uuid:8b3e8e2b-2e92-498c-851c-a1288c8eb25a",
  "dwc:basisOfRecord": "HumanObservation",
  "dwc:scientificName": "Bombus terrestris (Linnaeus, 1758)",
  "dwc:kingdom": "Animalia",
  "dwc:taxonRank": "species",
  "dwc:decimalLatitude": 50.0755,
  "dwc:decimalLongitude": 14.4378,
  "dwc:geodeticDatum": "WGS84",
  "dwc:coordinateUncertaintyInMeters": 15,
  "dwc:eventDate": "2024-06-12T10:15:00Z",
  "dwc:recordedBy": "Michal Torma",
  "dcterms:license": "http://creativecommons.org/publicdomain/zero/1.0/"
}
```

Mandatory Core Terms

- `occurrenceID`
- `basisOfRecord`
- `scientificName`
- `eventDate`

Strongly Recommended

- `decimalLatitude`
- `decimalLongitude`
- `geodeticDatum`
- `coordinateUncertaintyInMeters`

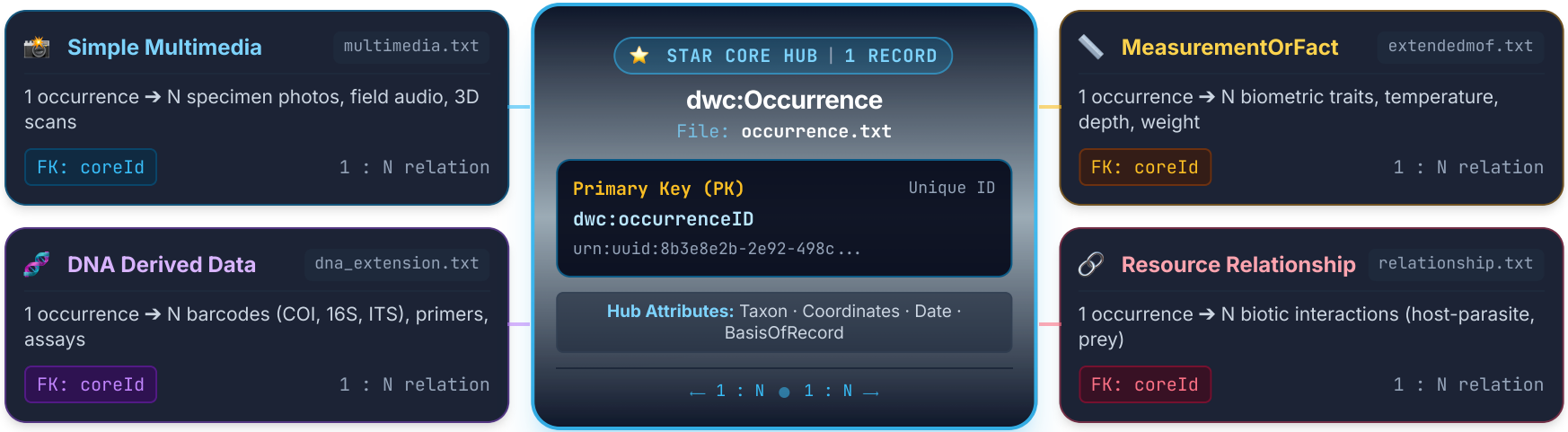
Mandatory Open License

Machine-readable license URL is required for GBIF publication:

- `CC0 1.0`
- `CC-BY 4.0`
- `CC-BY-NC 4.0`

The Darwin Core Star Schema (DwC-A)

A single flat table cannot cleanly model **one-to-many (1:N)** biological relationships. A **Darwin Core Archive (DwC-A)** solves this using a **Star Schema**:



Darwin Core Archive (.zip): Tied together by `meta.xml` (maps foreign keys `coreId → occurrenceID`) and `eml.xml` (dataset metadata & citation).

★ 1 Core : N Extensions

GBIF Extensions in Practice

GBIF maintains a community-curated Extension Registry to extend Core records with specialized data types:

📸 Simple Multimedia (Audubon Core)

Media

Attaches photos, field audio recordings, or 3D specimen scans to an occurrence record.

Terms: `identifier`, `format`, `license`

iNaturalist · Herbarium scans

🔪 Extended MeasurementOrFact (eMoF)

Traits & Sensors

Records quantitative traits (body length, mass) and environmental metrics (water temp, depth).

Terms: `measurementType`, `Value`, `Unit`

OBIS · Trait databases

🧬 DNA Derived Data (GGBN / MIxS)

eDNA & Sequences

Links DNA barcoding, metabarcoding, and environmental DNA (eDNA) detections to vouchers.

Terms: `target_gene`, `pcr_primer`, `accession`

NCBI · BOLD · ENA

🔗 Resource Relationship

Interactions

Documents ecological interactions between occurrences (host-parasite, pollinator, prey).

Terms: `relationshipOfResource`, `relatedResourceID`

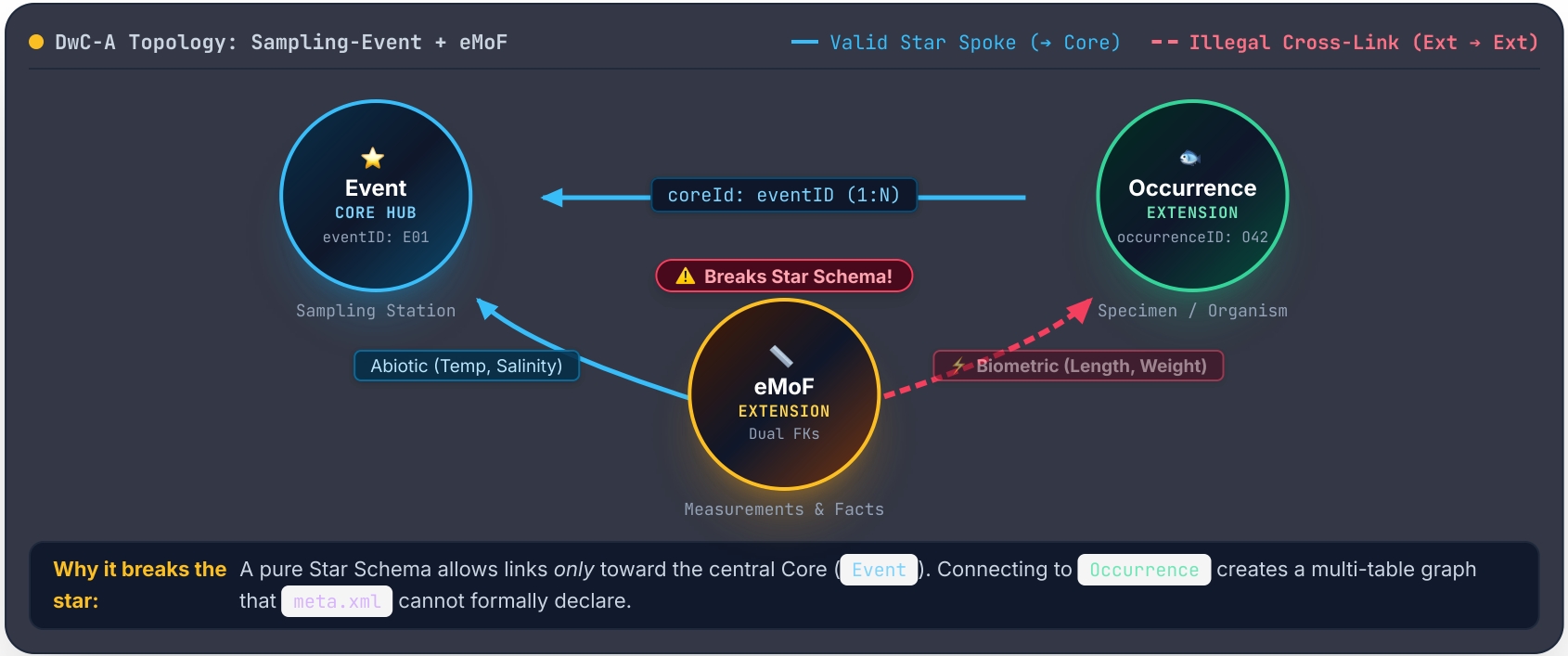
GloBI · Food webs

💡 Each extension file links back to `coreId` in `occurrence.txt` via schema defined in `meta.xml`

Explore Registry ↗

Special Case: ExtendedMeasurementOrFact (eMoF)

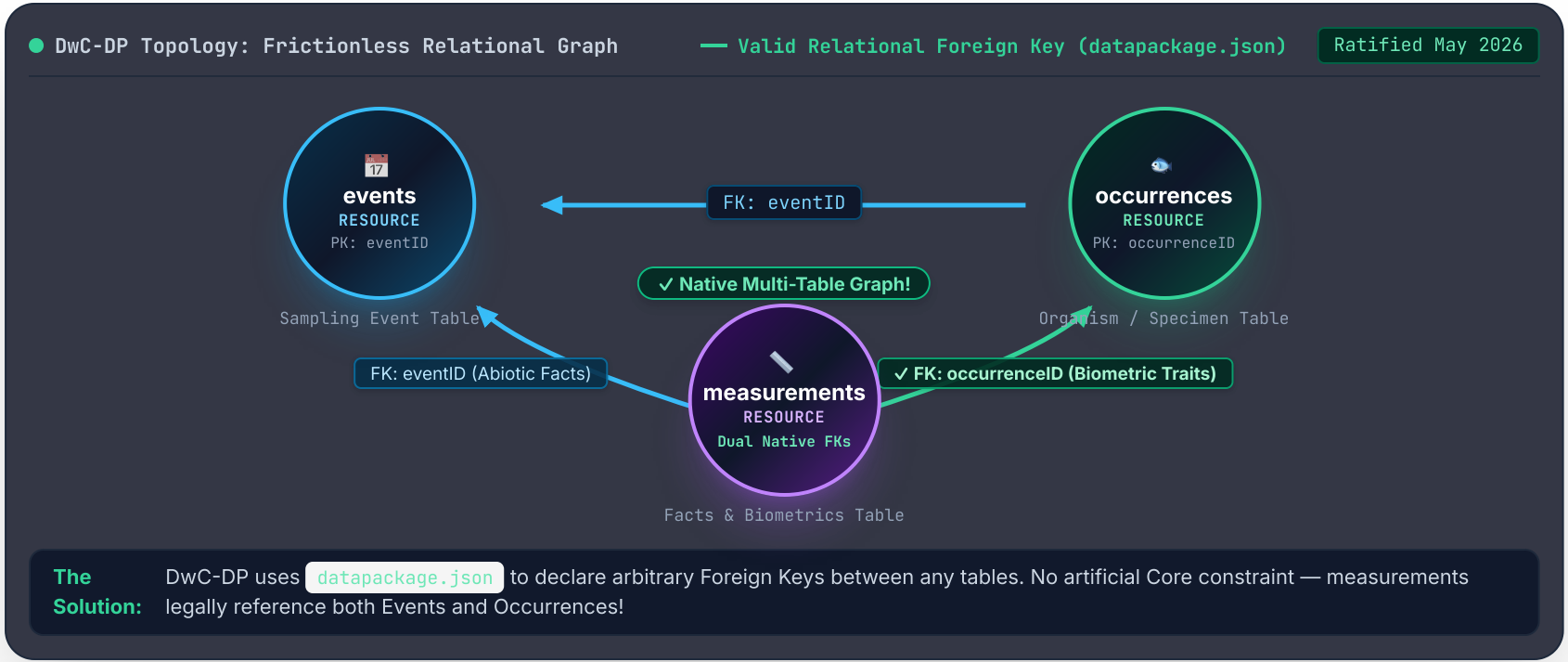
In Sampling-Event datasets, measurements target **both** the Event (abiotic) and Occurrence (biometric traits), breaking the pure Star Schema:



Why it breaks the star: A pure Star Schema allows links *only* toward the central Core (`Event`). Connecting to `Occurrence` creates a multi-table graph that `meta.xml` cannot formally declare.

The Solution: Darwin Core Data Package (DwC-DP)

Officially ratified by **TDWG on May 26, 2026**: Replacing the rigid Star Schema with Frictionless Data Packages and multi-table relational graphs.



The Solution: DwC-DP uses `datapackage.json` to declare arbitrary Foreign Keys between any tables. No artificial Core constraint — measurements legally reference both Events and Occurrences!